

transfer learning

stats403_deep_learning
spring_2025
lecture_12

12.1 overview of transfer learning

difficulty in machine learning

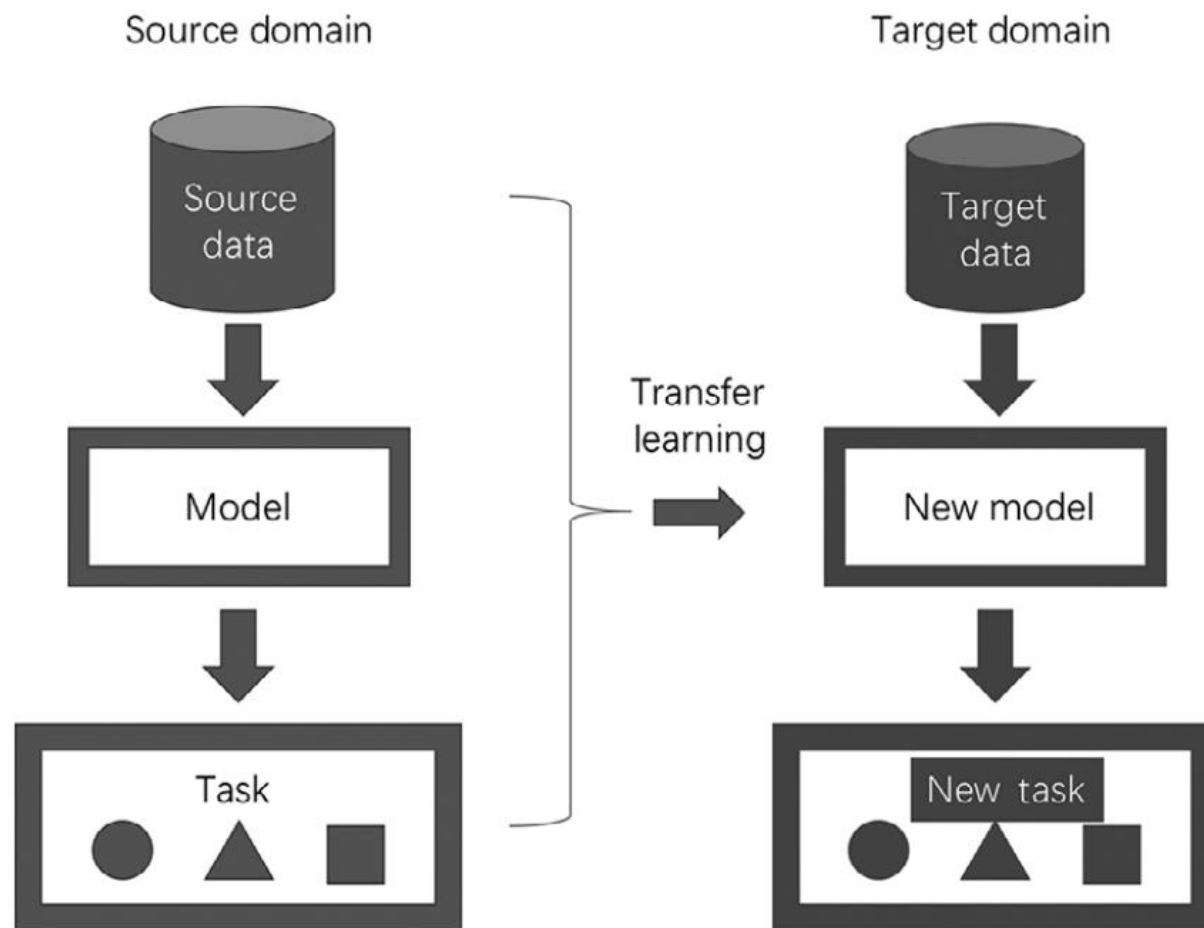
- a major challenge in practicing machine learning in many applications is that models do not work well in new task domains
- it would be impossible to collect large quantities of data in every domain before applying machine learning

call for transfer learning

1. many applications only have small data
2. machine learning models need to be robust
3. personalization and specialization are important issues
4. user privacy and data security are important issues

Transfer learning refers to the machine learning paradigm in which an algorithm extracts knowledge from one or more application scenarios to help boost the learning performance in a target scenario.

transfer learning



definition

- a domain $\mathbb{D}$ consists of two components: a feature space $\mathcal{X}$ and a marginal probability distribution $\mathbb{P}^X$, where each input instance $\mathbf{x} \in \mathcal{X}$
- in general, if two domains are different, then they may have different feature spaces or different marginal probability distributions

definition

- given a specific domain, $\mathbb{D} = \{\mathcal{X}, \mathbb{P}^X\}$, a task $\mathbb{T}$ consists of two components: a label space $\mathcal{Y}$ and a function $f(\cdot)$
- the task is denoted by $\mathbb{T} = \{\mathcal{Y}, f(\cdot)\}$
- the function $f(\cdot)$ is a predictive function that can be used to make predictions on unseen instances
- from a probabilistic viewpoint, $f(\mathbf{x})$ can be written as $P(y|\mathbf{x})$

definition

- source domain: $\mathbb{D}_s$
- target domain: $\mathbb{D}_t$
- source domain labeled data: $\mathcal{D}_s = \{(\mathbf{x}_{s_i}, y_{s_i})\}_{i=1}^{n_s}$
 $\mathbf{x}_{s_i} \in \mathcal{X}_s \quad y_{s_i} \in \mathcal{Y}_s$
- target domain labeled data: $\mathcal{D}_t = \{(\mathbf{x}_{t_i}, y_{t_i})\}_{i=1}^{n_t}$
 $\mathbf{x}_{t_i} \in \mathcal{X}_t \quad y_{t_i} \in \mathcal{Y}_t$
 $0 \leq n_t \ll n_s$

definition

- given a source domain $\mathbb{D}_s$ and learning task $\mathbb{T}_s$, a target domain $\mathbb{D}_t$ and learning task $\mathbb{T}_t$, **transfer learning** aims to help improve the learning of the target predictive function $f_t(\cdot)$ for the target domain using the knowledge in $\mathbb{D}_s$ and $\mathbb{T}_s$, where $\mathbb{D}_s \neq \mathbb{D}_t$ or $\mathbb{T}_s \neq \mathbb{T}_t$

definition

- given a source domain $\mathbb{D}_s$ and learning task $\mathbb{T}_s$, a target domain $\mathbb{D}_t$ and learning task $\mathbb{T}_t$, **transfer learning** aims to help improve the learning of the target predictive function $f_t(\cdot)$ for the target domain using the knowledge in $\mathbb{D}_s$ and $\mathbb{T}_s$, where $\mathbb{D}_s \neq \mathbb{D}_t$ or $\mathbb{T}_s \neq \mathbb{T}_t$

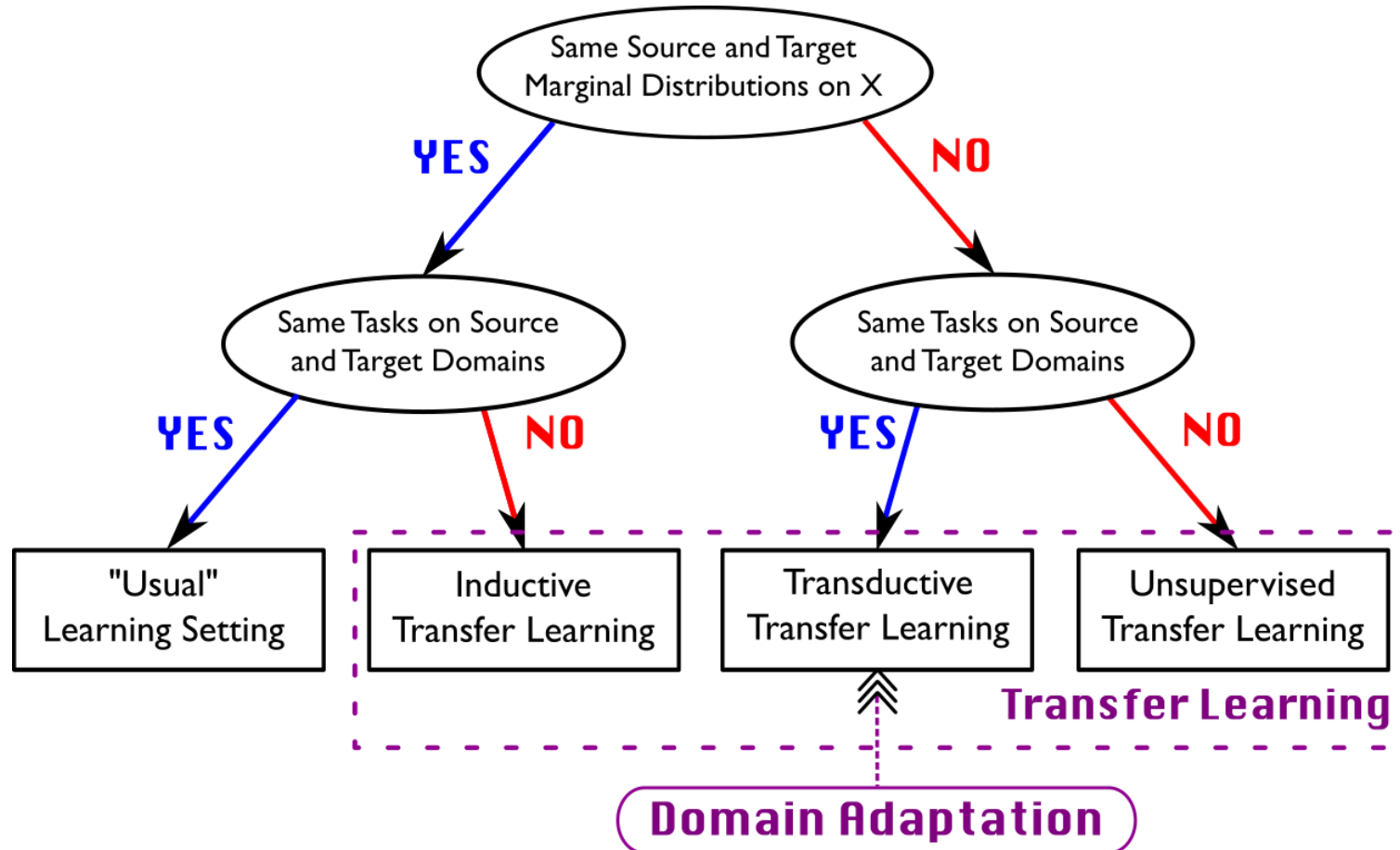
$$\mathbb{D}_s \neq \mathbb{D}_t \left\{ \begin{array}{l} \mathcal{X}_s \neq \mathcal{X}_t \\ \mathbb{P}^{X_s} \neq \mathbb{P}^{X_t} \end{array} \right.$$

$$\mathbb{T}_s \neq \mathbb{T}_t \left\{ \begin{array}{l} \mathcal{Y}_s \neq \mathcal{Y}_t \\ \mathbb{P}^{Y_s|X_s} \neq \mathbb{P}^{Y_t|X_t} \end{array} \right.$$

taxonomy

- homogeneous transfer learning: common instances exist and same label set
 - heterogeneous transfer learning: no common instance or different label sets
-
- supervised transfer learning: sufficiently many labeled data in target domain
 - semi-supervised transfer learning: few labeled data in target domain
 - unsupervised transfer learning: no labeled data in target domain

taxonomy

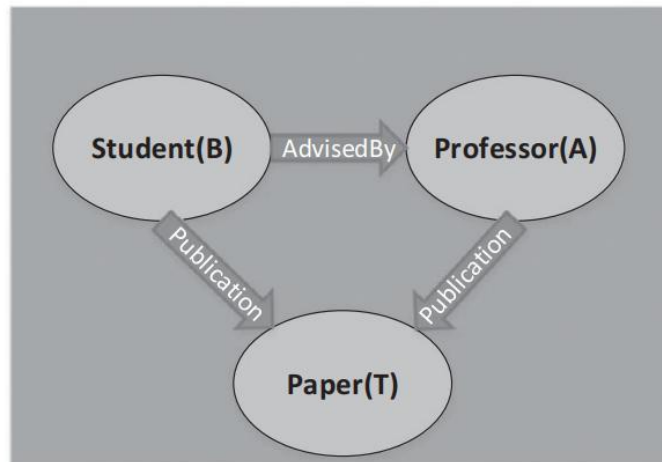


how to transfer

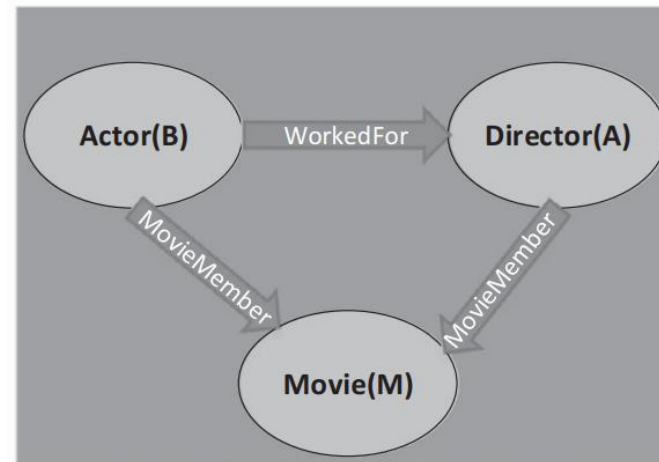
- instance-based algorithms
 - the knowledge transferred corresponds to the weights attached to source instances
- feature-based algorithms
 - the knowledge transferred corresponds to the subspace spanned by the features in the source and target domains
- model-based algorithms
 - the knowledge to be transferred is embedded in part of the source domain models
- relation-based algorithms
 - the knowledge to be transferred corresponds to rules specifying the relations between the entities in the source domain

relation-based

Academic domain (source)



Movie domain (target)



$AdvisedBy(B, A) \wedge Publication(B, T)$
 $\Rightarrow Publication(A, T)$

First-order transfer

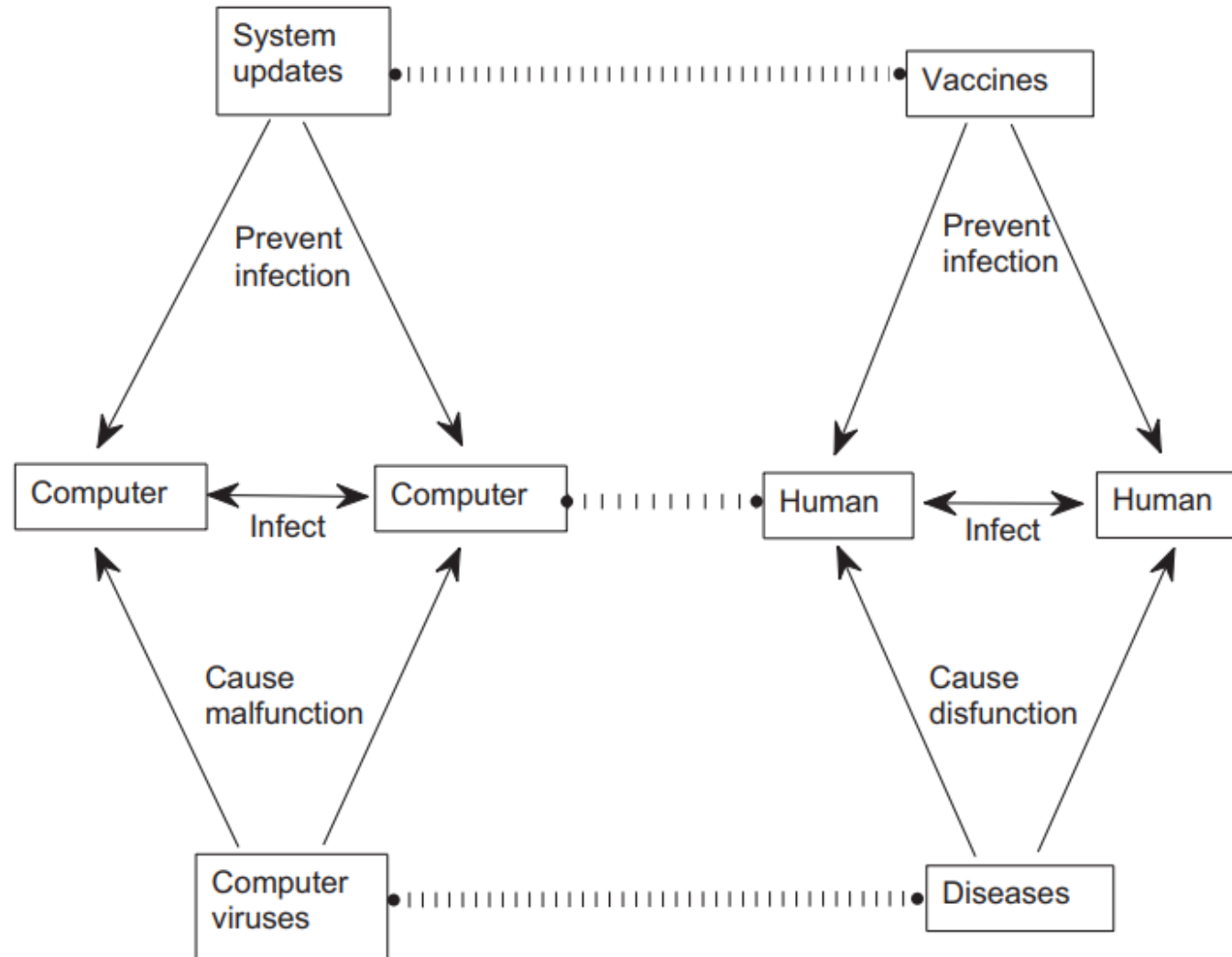
$WorkedFor(B, A) \wedge MovieMember(B, M)$
 $\Rightarrow MovieMember(A, M)$

Second-order transfer

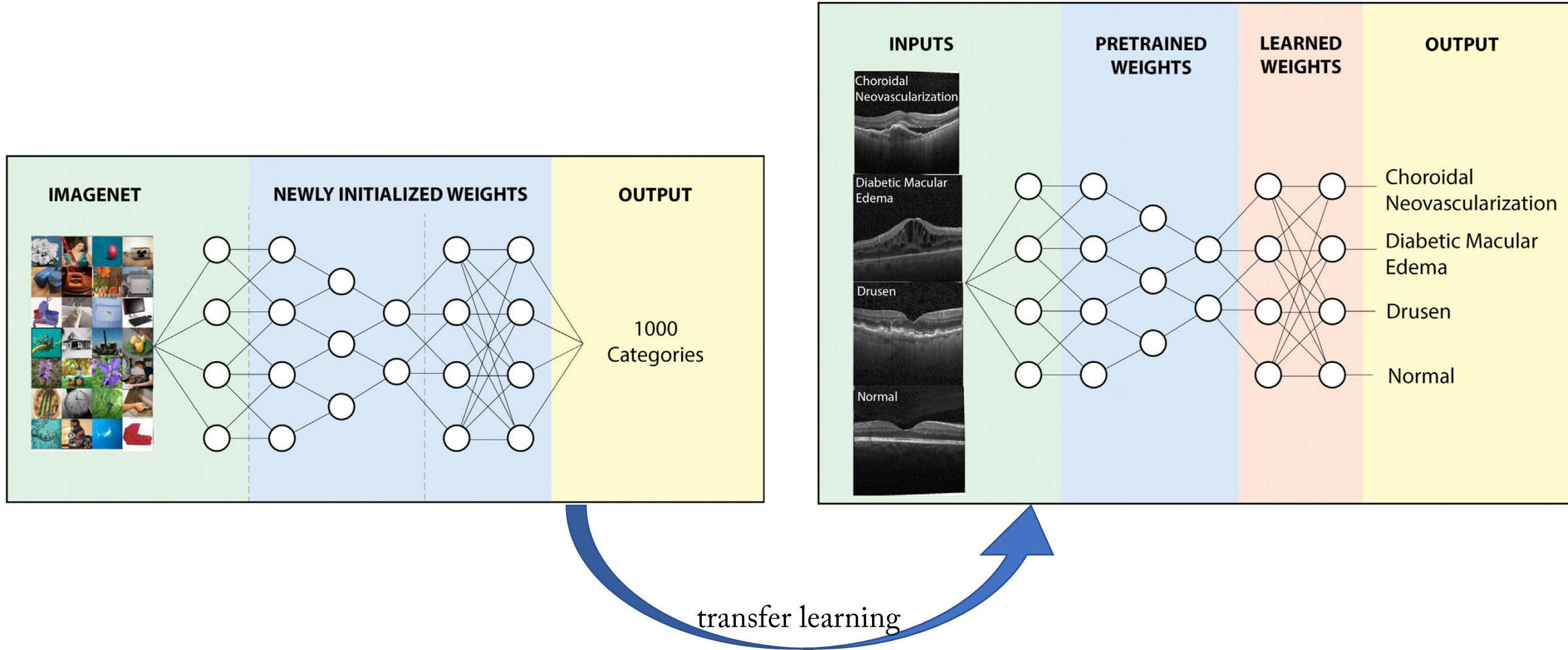
Second-order transfer

$P1(x, y) \wedge P2(x, z) \Rightarrow P2(y, z)$

relation-based

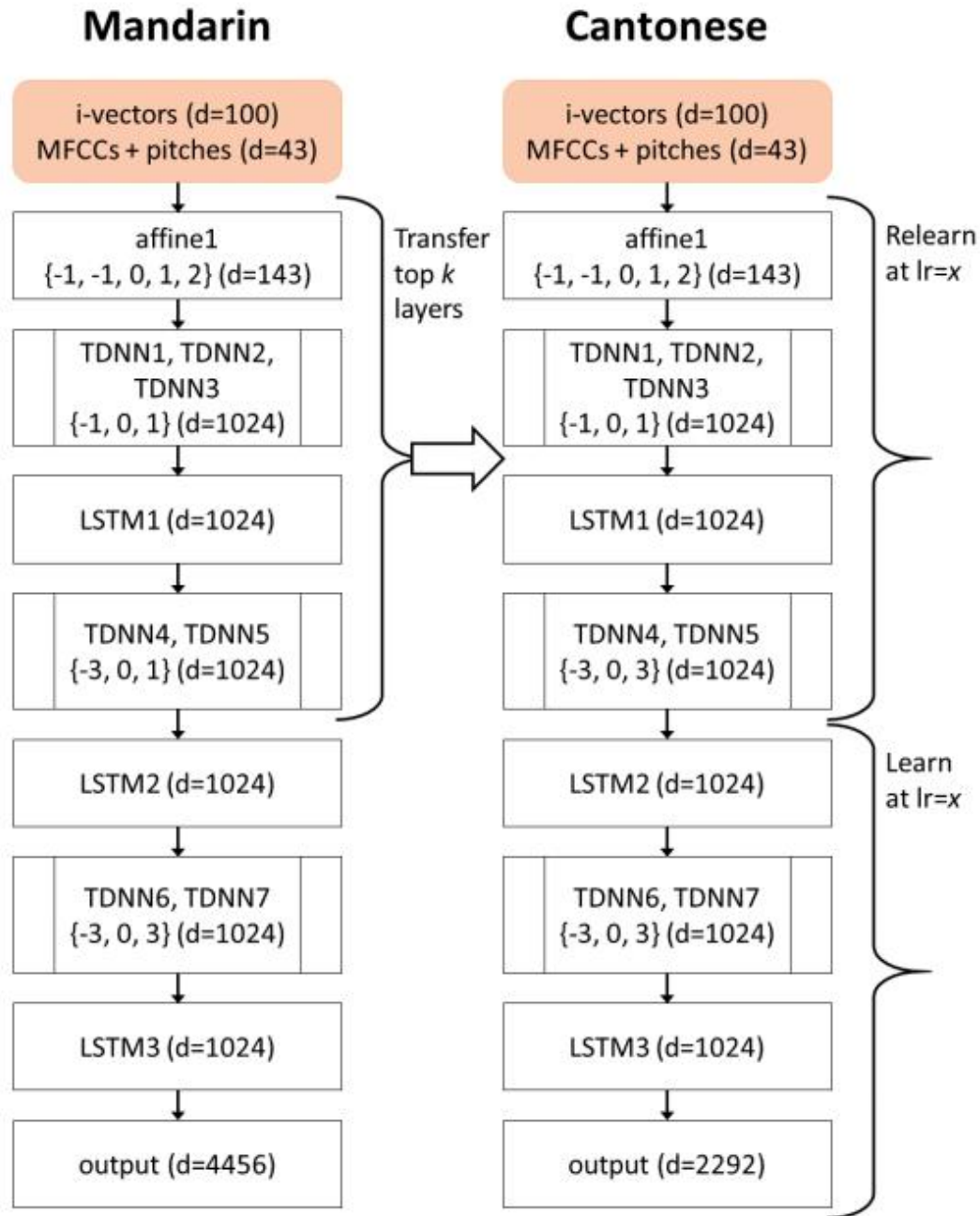


deep model-based



Kermany, D. S., Goldbaum, M., Cai, W., Valentim, C. C., Liang, H., Baxter, S. L., ... & Zhang, K. (2018). Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 172(5), 1122-1131.

deep model based



Li, B., Wang, X., & Beigi, H. (2019). Cantonese automatic speech recognition using transfer learning from mandarin. *arXiv preprint arXiv:1911.09271*.

transferability of CNN

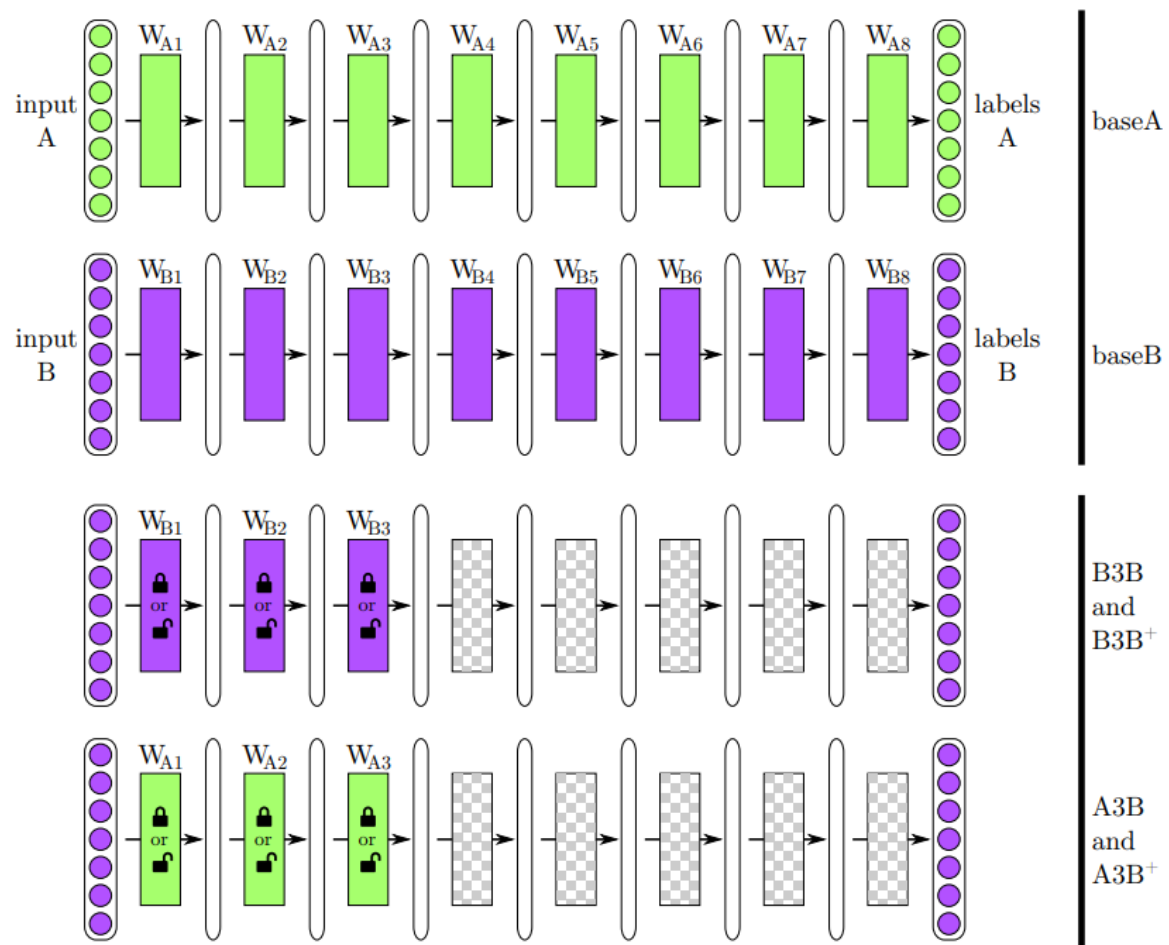


image credit: Yosinski, J., Clune, J., Bengio, Y., & Lipson, H. (2014). How transferable are features in deep neural networks?. In *NeurIPS*.

transferability of CNN

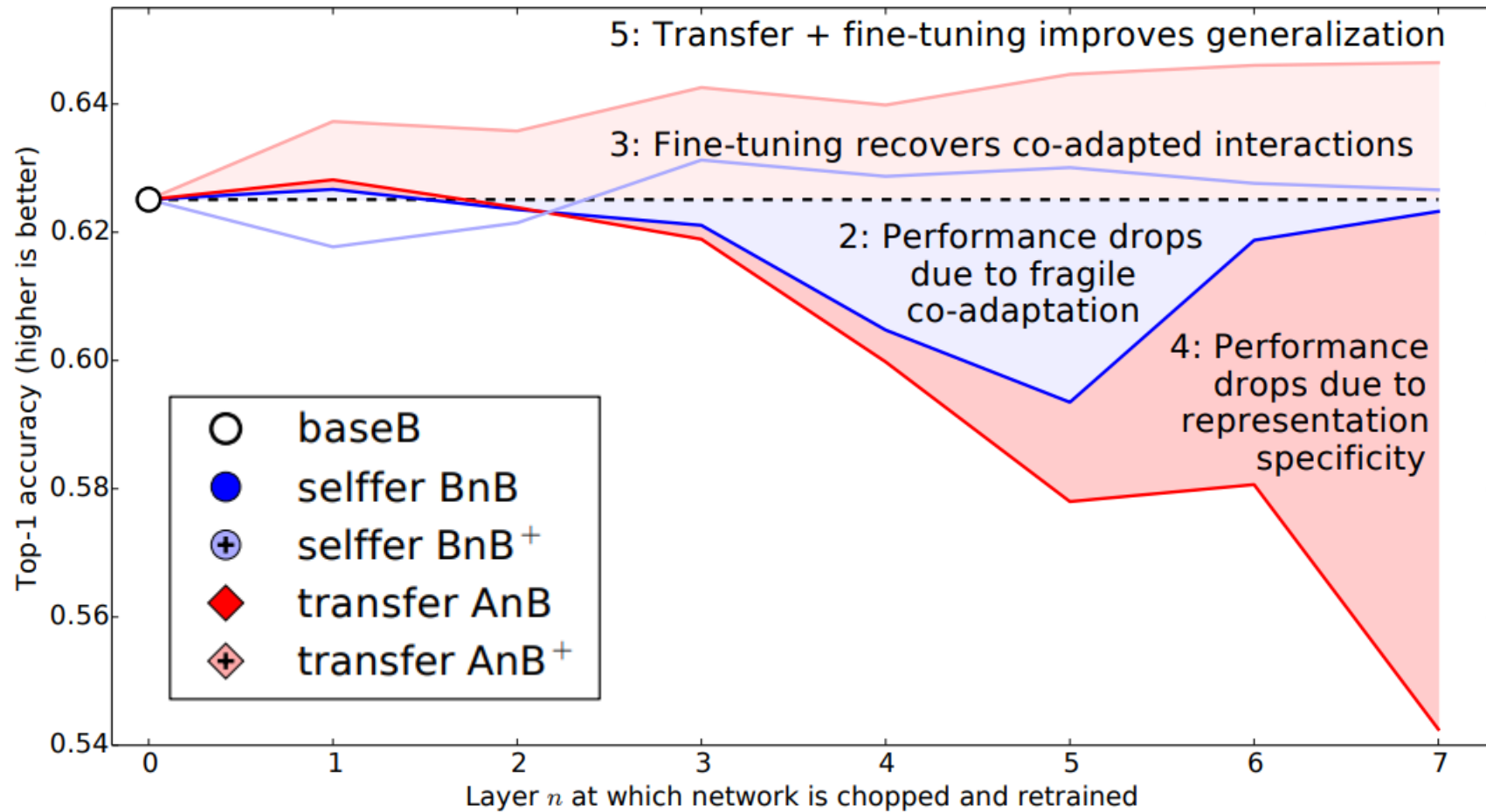
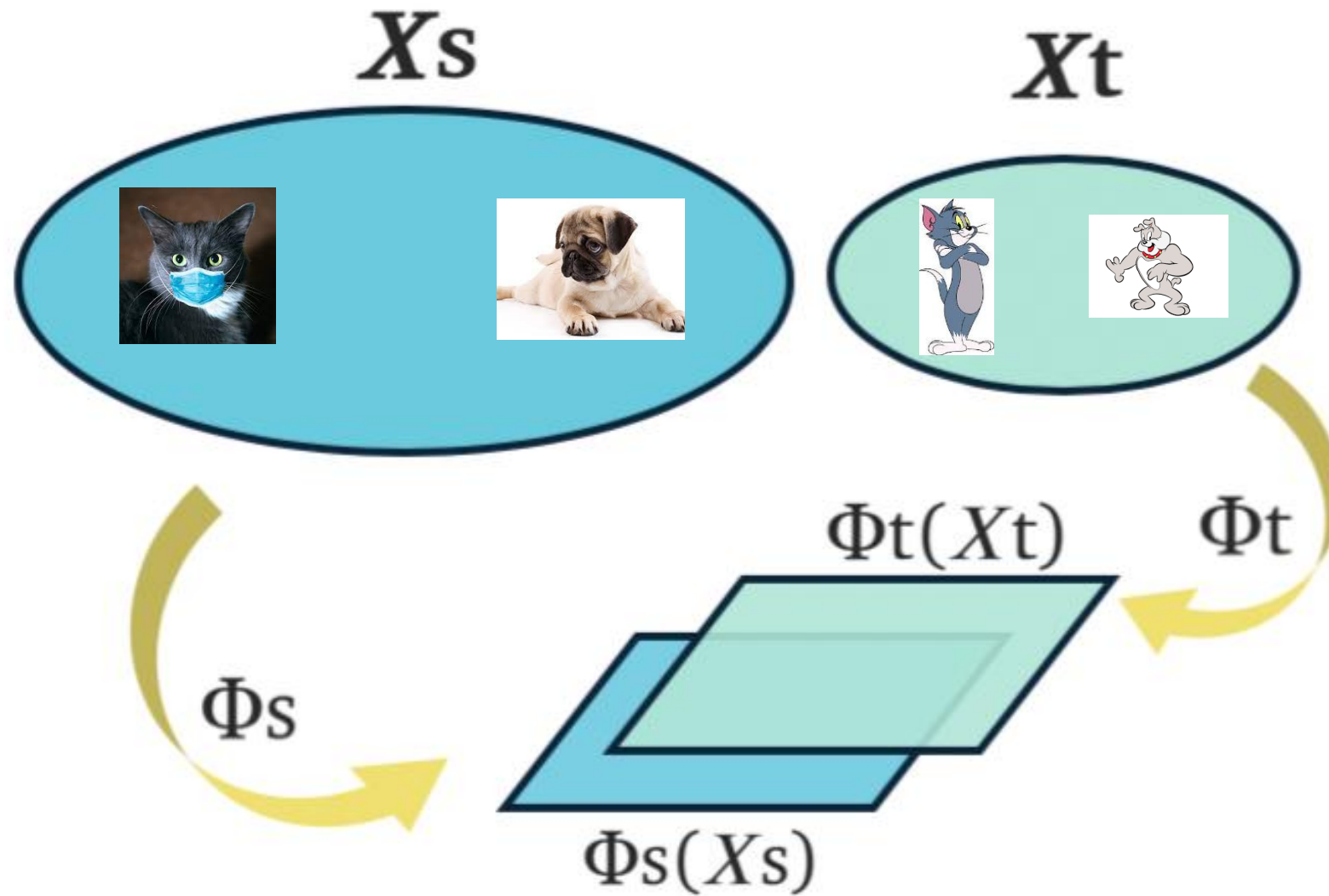


image credit: Yosinski, J., Clune, J., Bengio, Y., & Lipson, H. (2014). How transferable are features in deep neural networks?. In *NeurIPS*.

12.2 feature-based transfer learning

feature-based transfer learning



feature-based transfer learning

- idea: learn a pair of mapping functions $\{\varphi_s(\cdot), \varphi_t(\cdot)\}$ to map data respectively from the source domain and the target domain to a **common feature space**, where the difference between domains can be reduced.
- such feature space is said to be **domain-invariant** and a key issue in learning such domain invariant features is how to measure “domain invariance”

Maximum Mean Discrepancy (MMD)

- the **Maximum Mean Discrepancy (MMD)** between two domain samples is defined to be

$$\text{MMD}(\mathbf{X}_s, \mathbf{X}_t) = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(x_i^s) - \frac{1}{n_t} \sum_{i=1}^{n_t} \phi(x_i^t) \right\|_{\mathcal{H}}$$

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$$\text{MMD}^2(\mathbf{X}_s, \mathbf{X}_t) = \text{tr}(\mathbf{KL})$$

$$\mathbf{K} = \begin{bmatrix} \mathbf{K}_{s,s} & \mathbf{K}_{s,t} \\ \mathbf{K}_{s,t}^T & \mathbf{K}_{t,t} \end{bmatrix} \in \mathbb{R}^{(n_s+n_t) \times (n_s+n_t)} \quad l_{ij} = \begin{cases} \frac{1}{n_s^2} & x_i, x_j \in X_s \\ \frac{1}{n_t^2} & x_i, x_j \in X_t \\ -\frac{1}{n_s n_t} & \text{otherwise.} \end{cases}$$

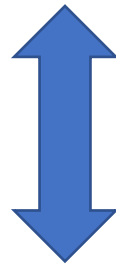
Maximum Mean Discrepancy (MMD)

- side note:

$$\text{MMD}(P, Q) = \sup_{\|f\|_{\mathcal{H}} \leq 1} (\mathbb{E}[f(X)] - \mathbb{E}[f(Y)])$$

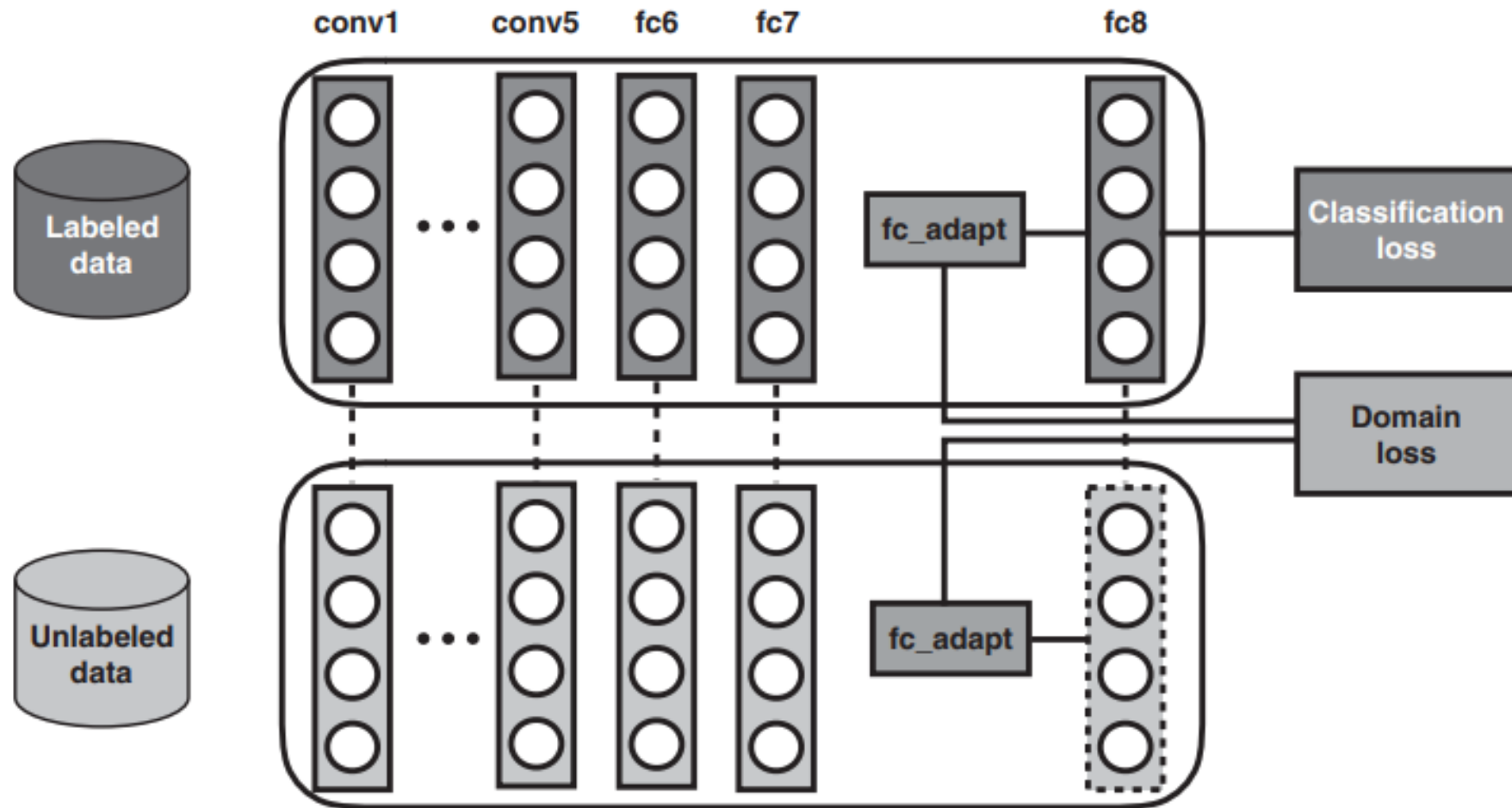
MMD embedding

$$\begin{aligned} \min_{\varphi} \quad & \text{MMD}(\varphi(\mathbf{X}_S), \varphi(\mathbf{X}_T)) + \lambda \Omega(\varphi) \\ \text{s.t.} \quad & \text{constraints on } \varphi(\mathbf{X}_S) \text{ and } \varphi(\mathbf{X}_T) \end{aligned}$$

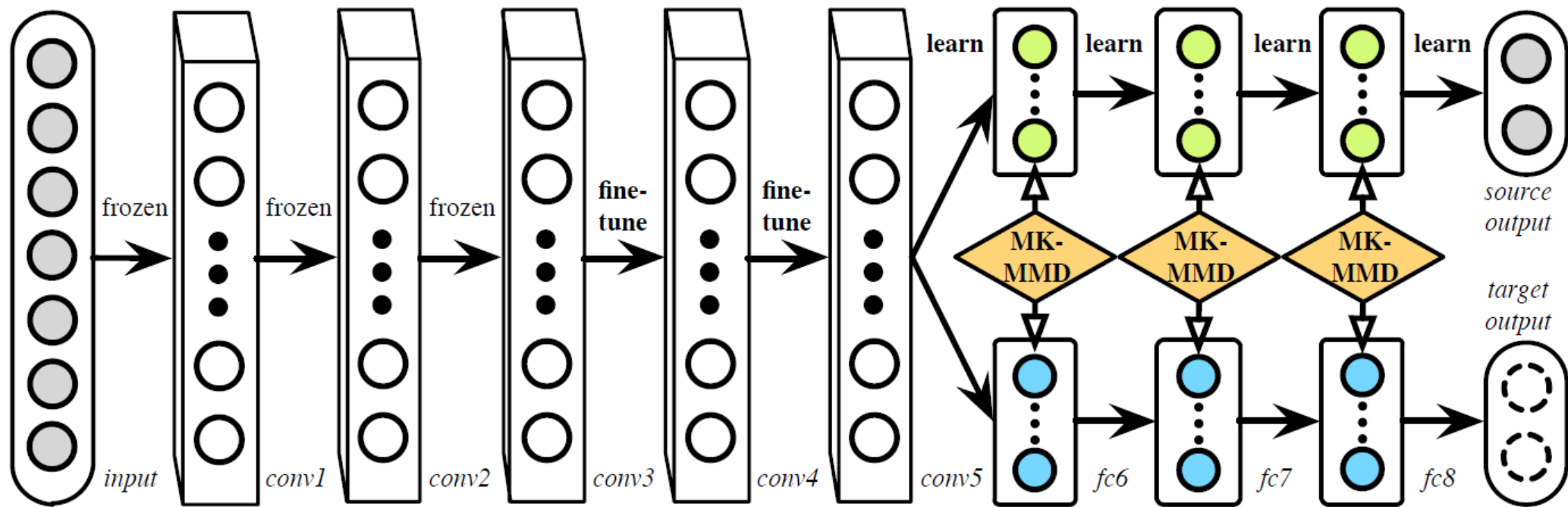


$$\begin{aligned} \min_{\varphi} \quad & \text{tr}(\mathbf{KL}) + \lambda \Omega(\varphi) \\ \text{s.t.} \quad & \text{constraints on } \varphi(\mathbf{X}_S) \text{ and } \varphi(\mathbf{X}_T) \end{aligned}$$

deep architectures with MMD



deep architectures with MMD



analysis of transfer learning

- setting:

$$\mathcal{X}_s = \mathcal{X}_t = \mathcal{X} \subset \mathbb{R}^d$$

$$\mathcal{Y}_s = \mathcal{Y}_t = \mathcal{Y} = \{0, 1\}$$

$$\mathbb{P}^{X_s} = P, \mathbb{P}^{X_t} = Q$$

$$\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$$

$$\mathcal{D}_t = \{(x_i^t, ?)\}_{i=1}^{n_t}$$

true labeling function $f : \mathcal{X} \rightarrow \{0, 1\}$

analysis of transfer learning

Definition ($\mathcal{H}$ -divergence) Given two distributions P and Q . Let $\mathcal{H}$ be the hypothesis class. Also let

$$\mathbf{x} \in \mathbb{I}(h) \Leftrightarrow h(\mathbf{x}) = 1$$

We define the $\mathcal{H}$ -divergence between P and Q as

$$d_{\mathcal{H}}(P, Q) = 2 \sup_{h \in \mathcal{H}} |\mathbb{P}_P \mathbb{I}(h) - \mathbb{P}_Q \mathbb{I}(h)|$$

analysis of transfer learning

- Define

$$\varepsilon_s(h) = \mathbb{E}_P[f(\mathbf{x}) \neq h(\mathbf{x})]$$

$$\varepsilon_t(h) = \mathbb{E}_Q[f(\mathbf{x}) \neq h(\mathbf{x})]$$

and the empirical versions (expectation taken on data)

$$\hat{\varepsilon}_s(h), \hat{\varepsilon}_t(h)$$

analysis of transfer learning

Theorem Let $\mathcal{H}$ be a hypothesis class with VC-dimension d . Suppose m instances are i.i.d. sampled from the source domain. Also, let

$$h^* = \operatorname{argmin}_{h \in \mathcal{H}} \varepsilon_s(h) + \varepsilon_t(h)$$

and

$$\lambda = \varepsilon_s(h^*) + \varepsilon_t(h^*)$$

Then with probability at least $1 - \delta$, for any h in $\mathcal{H}$, we have

$$\varepsilon_t(h) \leq \hat{\varepsilon}_s(h) + d_{\mathcal{H}}(\mathcal{D}_s, \mathcal{D}_t) + \lambda + \sqrt{\frac{4}{m} \left(d \log \frac{2em}{d} + \log \frac{4}{\delta} \right)}$$

analysis of transfer learning

Proof: We would like to upper bound

$$\Sigma_t(h) = \mathbb{P}_Q(f \neq h) = \mathbb{P}_Q(\{x \in \mathcal{X} : f(x) \neq h(x)\})$$

Note that $\{x \in \mathcal{X} : f(x) \neq h(x)\}$

$$\subseteq \{x \in \mathcal{X} : h^*(x) \neq f(x) \text{ or } h^*(x) \neq h(x)\}$$

$$= \{x \in \mathcal{X} : h^*(x) \neq f(x)\} \cup \{x \in \mathcal{X} : h^*(x) \neq h(x)\}$$

analysis of transfer learning

Therefore,

$$\epsilon_t(h) = P_Q(f \neq h)$$

$$\leq P_Q(h^* \neq f) + P_Q(h^* \neq h)$$

$$\leq \epsilon_t(h^*) + P_P(h^* \neq h) + |P_Q(h^* \neq h) - P_P(h^* \neq h)|$$

$$\leq \epsilon_t(h^*) + P_P(h^* \neq h) + d_H(P, Q)$$

$$\leq \epsilon_t(h^*) + \underbrace{P_P(h^* \neq f)}_{\epsilon_S(h^*)} + \underbrace{P_P(h \neq f)}_{\epsilon_S(h)} + d_H(P, Q)$$

analysis of transfer learning

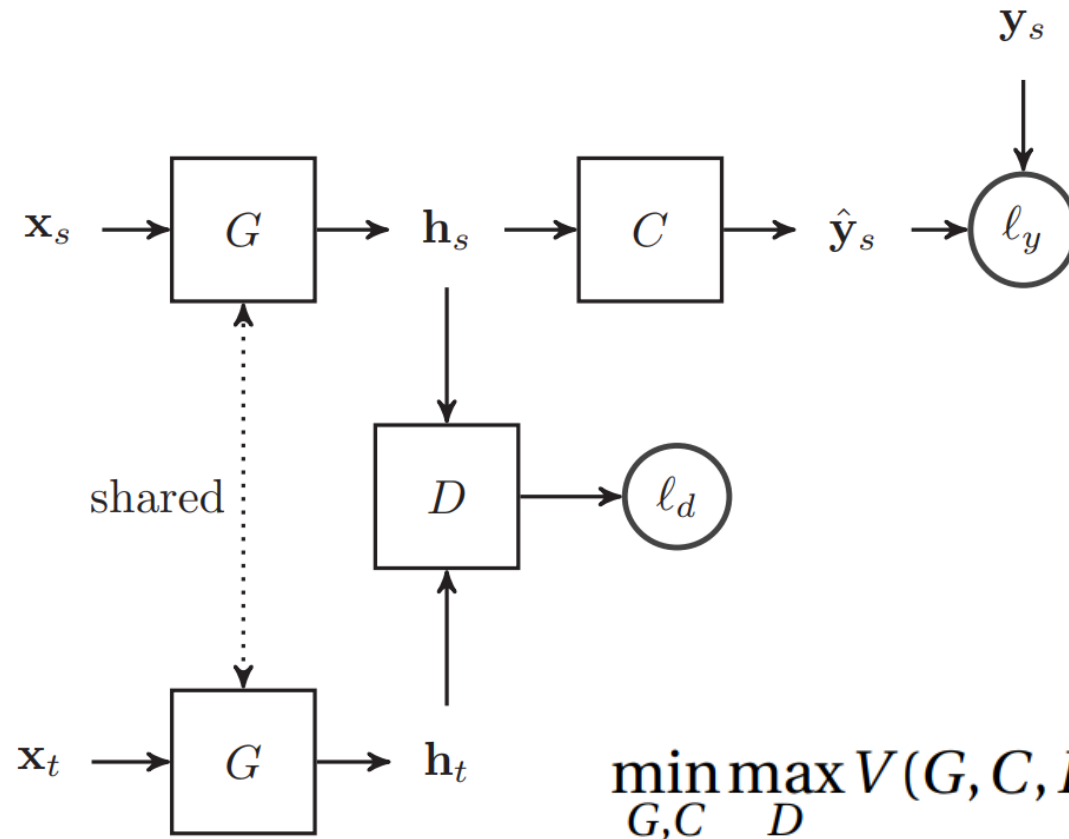
By PAC learning theory, with probability at least $1-\delta$

$$\xi_S(h) \leq \hat{\xi}_S(h) + \sqrt{\frac{4}{m} \left(d \log \frac{2em}{d} + \log \frac{4}{\delta} \right)}$$



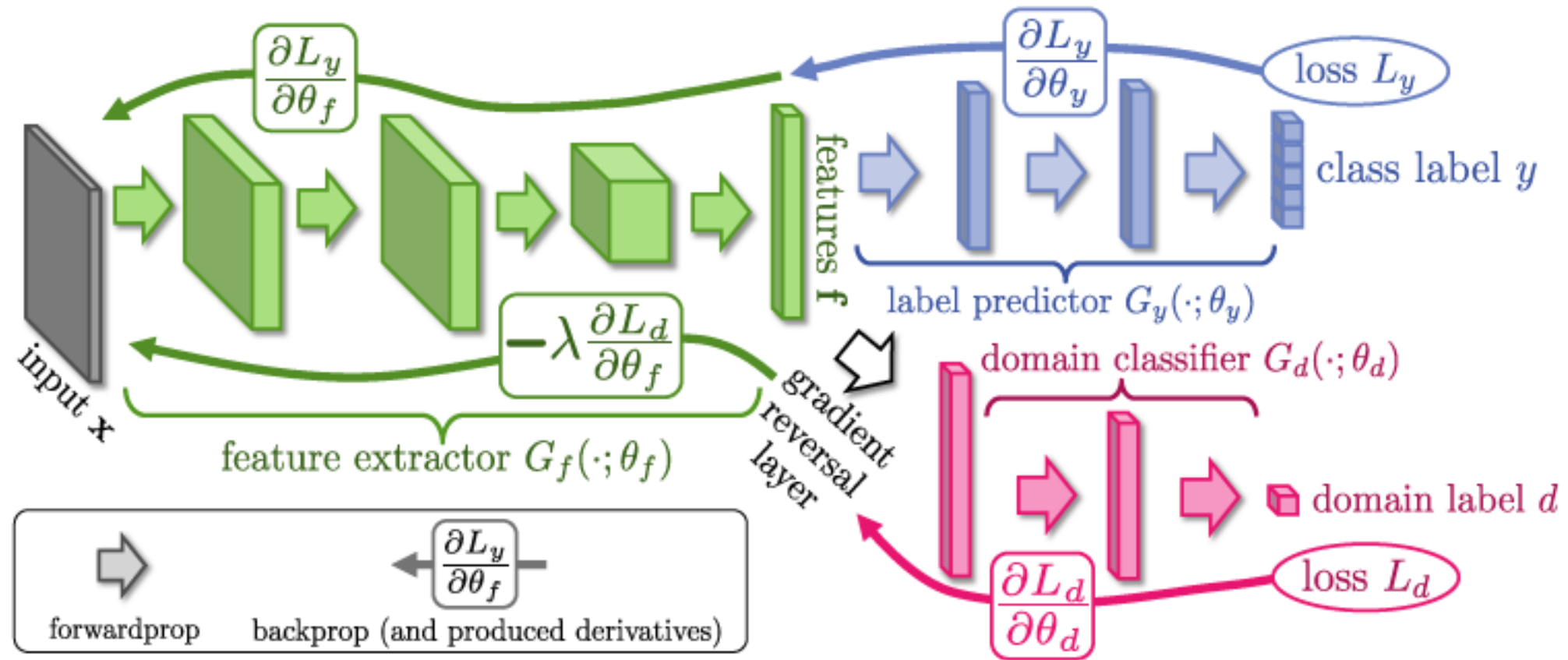
12.3 domain adversarial neural network (DANN)

domain adversarial neural network (DANN)



$$V(G, C, D) = \frac{1}{n_s} \sum_{i=1}^{n_s} \ell_y^i(G, C) - \lambda \left(\frac{1}{n_s} \sum_{i=1}^{n_s} \ell_d^i(G, D) + \frac{1}{n_t} \sum_{i=1}^{n_t} \ell_d^i(G, D) \right)$$

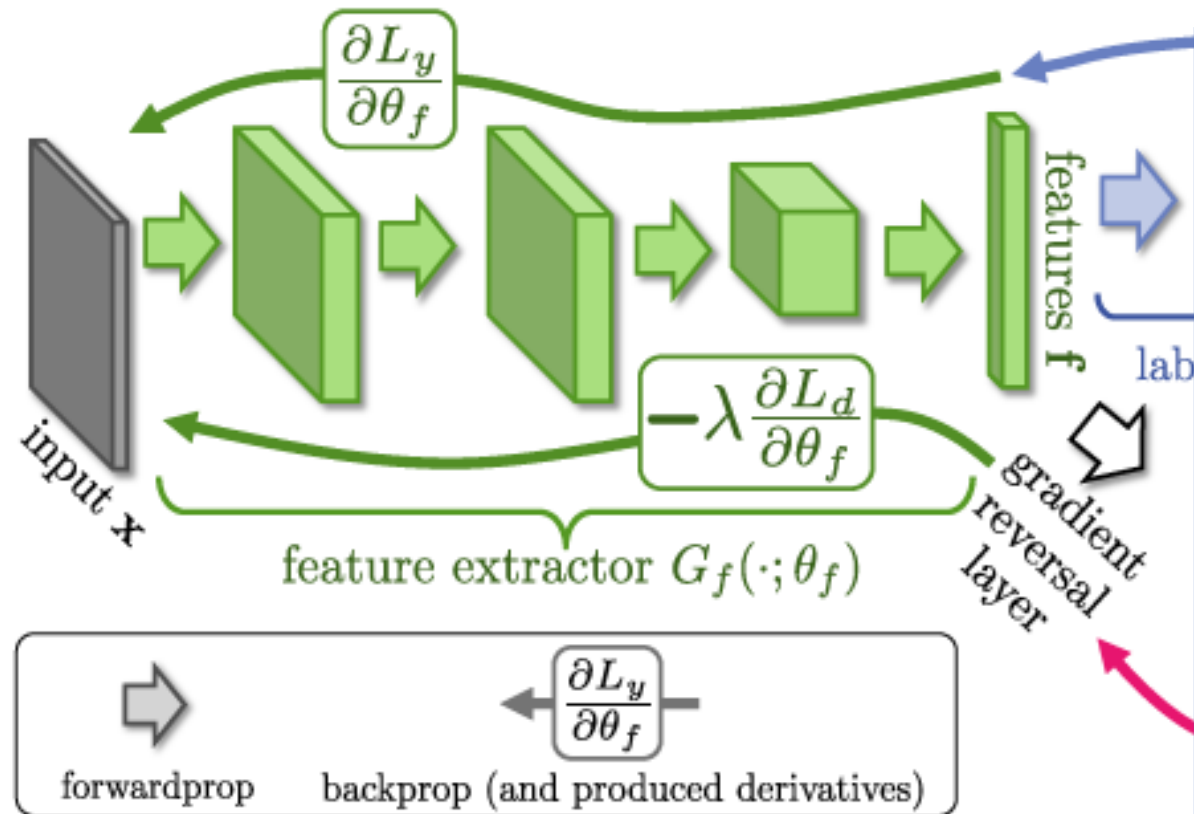
domain adversarial neural network (DANN)



domain adversarial neural network (DANN)

$$R_\lambda(\mathbf{x}) = \mathbf{x}$$

$$\frac{dR_\lambda}{d\mathbf{x}} = -\lambda \mathbf{I}$$



```
from torch.autograd import Function
```

```
class ReverseLayerF(Function):
```

```
    @staticmethod
```

```
    def forward(ctx, x, alpha):
```

```
        ctx.alpha = alpha
```

```
        return x.view_as(x)
```

```
    @staticmethod
```

```
    def backward(ctx, grad_output):
```

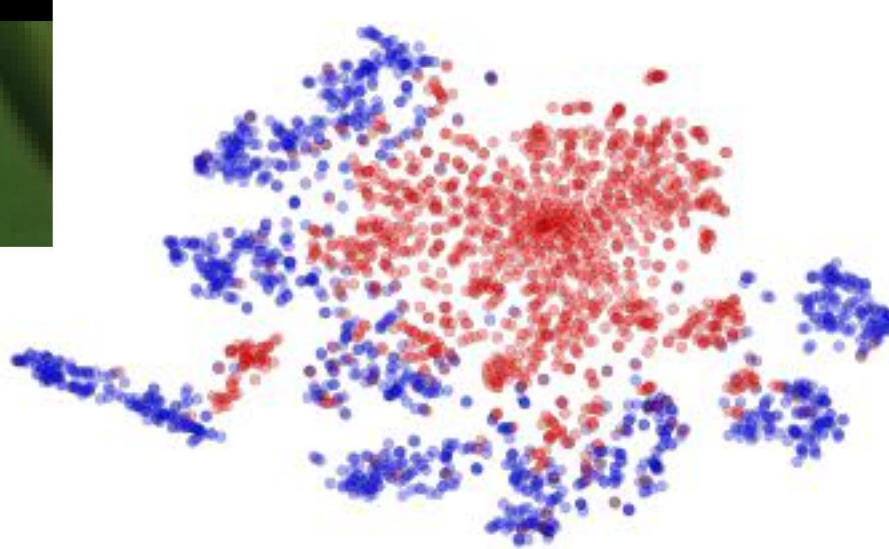
```
        output = grad_output.neg() * ctx.alpha
```

```
        return output, None
```

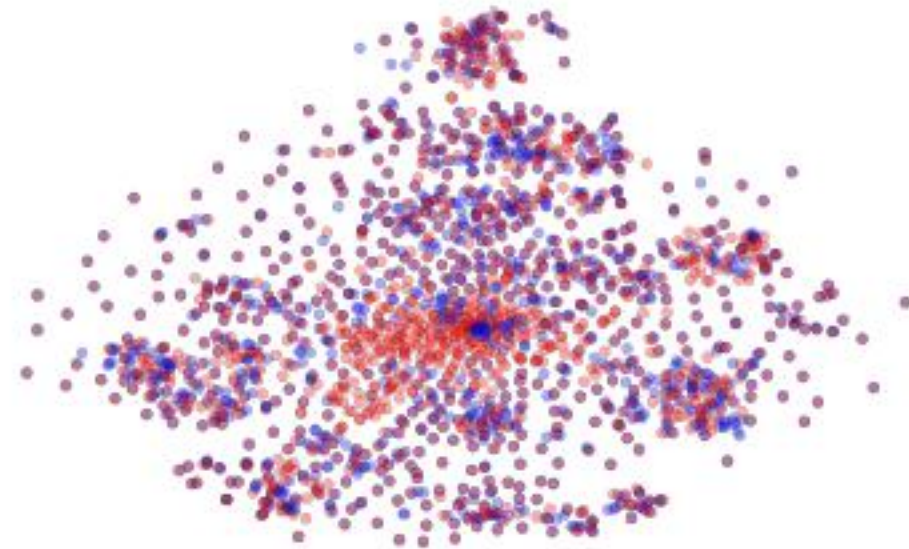
domain adversarial neural network (DANN)



MNIST $\rightarrow$ MNIST-M: top feature extractor layer

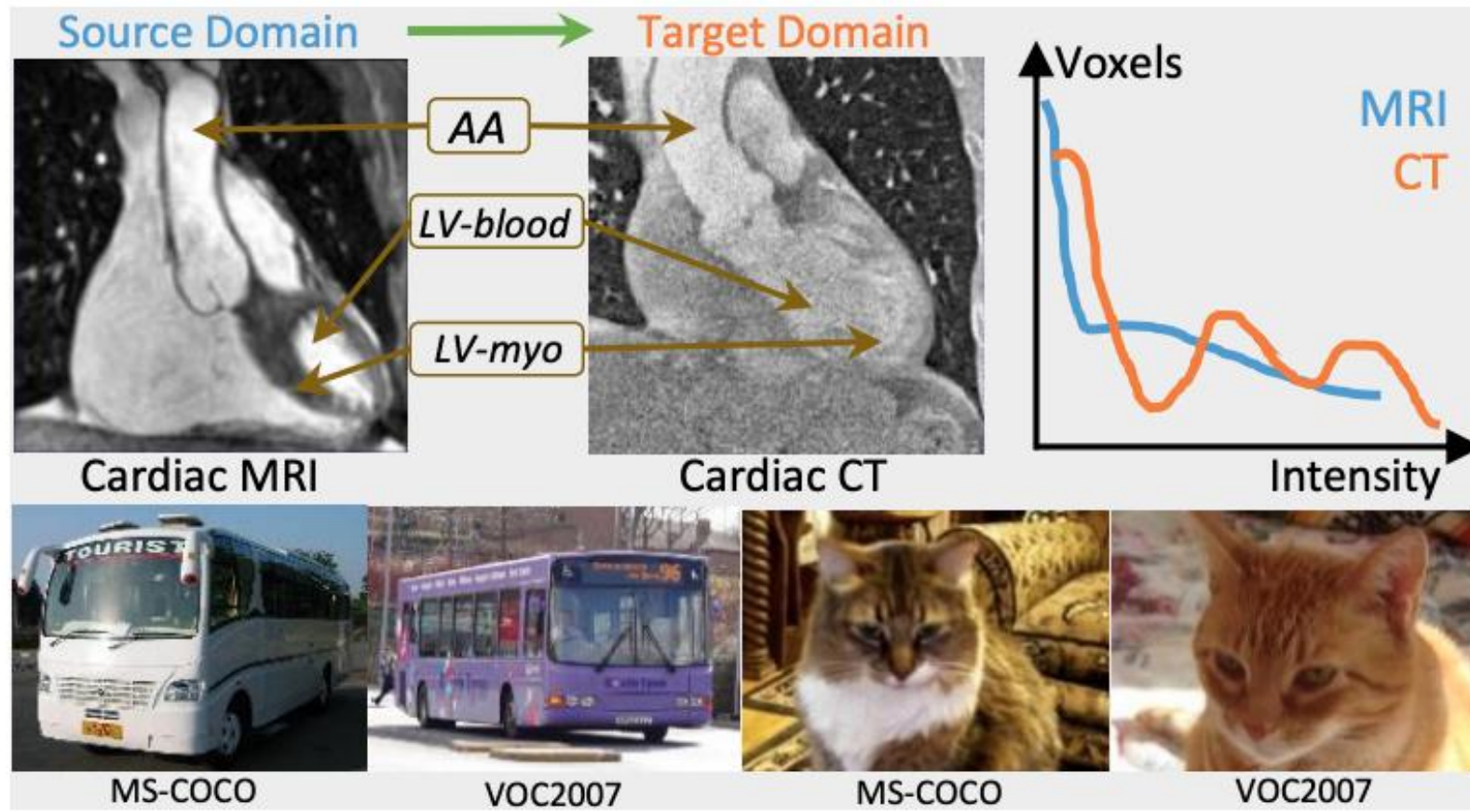


(a) Non-adapted



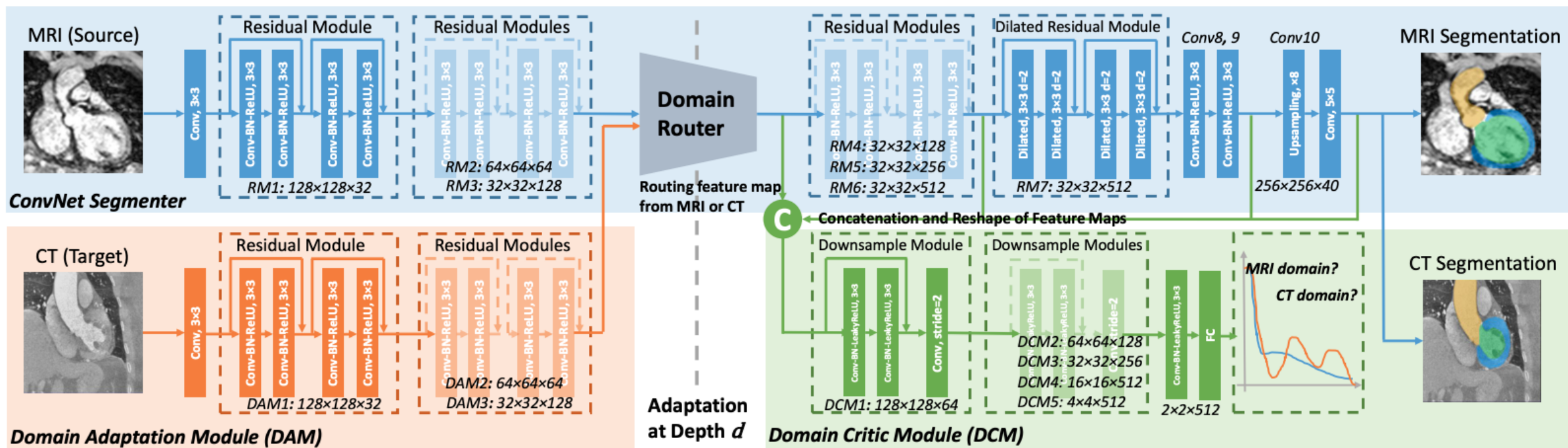
(b) Adapted

adversarial transfer



Dou, Q., Ouyang, C., Chen, C., Chen, H., & Heng, P. A. (2018). Unsupervised cross-modality domain adaptation of convnets for biomedical image segmentations with adversarial loss. In *IJCAI*.

adversarial transfer



Dou, Q., Ouyang, C., Chen, C., Chen, H., & Heng, P. A. (2018). Unsupervised cross-modality domain adaptation of convnets for biomedical image segmentations with adversarial loss. In *IJCAI*.

Thank you!

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